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Improved object recognition memory using post-encoding repetitive transcranial magnetic stimulation

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ABSTRACT

Background: Brain stimulation is known to affect canonical pathways and proteins involved in memory. However, there are conflicting results on the ability of brain stimulation to improve to memory, which may be due to variations in timing of stimulation.
Hypothesis: We hypothesized that repetitive transcranial magnetic stimulation (rTMS) given following a learning task and within the time period before retrieval could help improve memory.
Methods: We implanted male B6129SF2/J mice (n = 32) with a cranial attachment to secure the rTMS coil so that the mice could be given consistent stimulation to the frontal area whilst freely moving. Mice then underwent the object recognition test sampling phase and given treatment +3, +24, +48 h following the test. Treatment consisted of 10 min 10 Hz rTMS stimulation (TMS, n = 10), sham treatment (SHAM, n = 11) or a control group which did not do the behavior test or receive rTMS (CONTROL n = 11). At +72 h mice were tested for their exploration of the novel vs familiar object.
Results: At 72-h's, only the mice which received rTMS had greater exploration of the novel object than the familiar object. We further show that promoting synaptic GluR2 and maintaining synaptic connections in the perirhinal cortex and hippocampal CA1 are important for this effect. In addition, we found evidence that these changes were linked to CAMKII and CREB pathways in hippocampal neurons.
Conclusion: By linking the known biological effects of rTMS to memory pathways we provide evidence that rTMS is effective in improving memory when given during the consolidation and maintenance phases.
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1. Introduction

Non-invasive brain stimulation is known to affect canonical pathways and proteins involved in memory. However, current studies which have examined the effects of brain stimulation, specifically repetitive transcranial magnetic stimulation (rTMS), on memory have had conflicting results [1]. A reason for this could be from targeting the stimulation at different times during memory formation and retrieval [1]. Evidence suggests that targeting rTMS at the post-encoding or consolidation phases could prove beneficial for developing stable long-term memories [2]–[4].

Consolidation is the phase that transitions a short-term memory into a long-term memory (LTM). The process is dependent on time-specific protein synthesis, which transforms the new information into stable synaptic modifications with increased postsynaptic receptor densities [5,6]. The importance of this time- and region-specific protein synthesis has been shown in studies that block *de novo* protein synthesis. For example, one study which blocked protein synthesis in the hippocampus 3-h, but not 6-h post a learning event, impaired LTM but not short-term memory in rats [7]. Of these proteins, several important signaling pathways have been identified that are upregulated and important in this process.

Multiple signaling pathways exist at the synapse which signal the receptor innervation and spine stability needed for LTM. These are activated through a long-term potentiation (LTP) stimulus event and are often enhanced with neurotrophic factors such as brain

Abbreviations: repetitive transcranial magnetic stimulation, rTMS; long-term memory, LTM; long-term potentiation, LTP; Object Recognition Test, ORT; frontal cortex, FC; entorhinal cortex, EC; perirhinal cortex, PC.

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derived neurotrophic factor (BDNF) [8]. BDNF is of particular importance in creating long-term memories, as blocking the synthesis of BDNF at specific time points following learning blocks its consolidation [5]. BDNF binds to tyrosine kinase receptors (TrkB) and can initiate multiple signaling pathways such as mitogen-activated protein kinase/extracellular signal-regulated kinase (ERK) or phospholipase Cγ pathways. These signaling pathways converge on activating the transcription regulation factor cyclic AMP-responsive element-binding protein (CREB). Calcium activated proteins such as calcium/calmodulin-dependent protein kinase II (CAMKII) also play a role in CREB activated gene regulation [9–11] and they also can directly mediate the synaptic expression of the glutamatergic AMPA receptors and stabilize synaptic scaffolding [12–15]. Together, these signaling pathways result in a shift in receptor expression and translocation that mediates the increased activation at the synapse, thereby serving to stabilize the synapses and contribute to the maintenance of prolonged LTP [16]. Importantly, previous studies have shown it is possible to improve memory consolidation through periodic reactivation of these consolidation pathways in experimental settings [17].

Previous studies have shown the ability for rTMS to directly influence these specific pathways in the brain. *In vivo* studies have shown that rTMS increases BDNF levels in the brain [18–20]. More directly, *in vitro* rTMS results in increased Ca^{2+} in neurons, a predominant signaling molecule in the CAMKII pathway [18]. Increased CAMKII and CREB phosphorylation was also shown *in vitro* after rTMS [20,21]. rTMS also directly influences synaptic receptor expression, increasing AMPA receptor density [22–24], and regulating AMPA receptors subunits Glutamate Receptor 1 (GluR1) and Glutamate Receptor 2 (GluR2) [25,26]. Therefore, as rTMS activates these pathways it would be most effective in improving memory during the consolidation phase.

This study examined the use of rTMS for improving memory by targeting the stimulation to the consolidation and maintenance phases of memory. We hypothesized that rTMS administered following a learning task, and within the time period before retrieval, could improve memory performance. This would be a result of periodic reactivation of pathways that are involved in memory, which are also influenced by rTMS. Using an animal model of a specific episodic LTM, we showed that rTMS improved object memory when given within the 72-h between encoding and retrieval, whereas the mice given the SHAM treatment did not maintain the memory. This effect was supported by changes to synaptic connections in the perirhinal cortex and CA1, elevated activation of CREB and CAMKII and a dynamic shift in AMPA receptor subunits at the synapse.

2. Material and methods

2.1. Animals

We used n = 33 male B6129SF2/J mice aged 2–4 months for this study. All mice lived under a 12-h light/12-h dark cycle with free access to dry feed and water. All animal care and procedures complied with the Animal Welfare Act and were in accordance with institutional guidelines and approved by the VA Palo Alto Committee on Animal Research.

2.2. Coil support surgery

Each of the mice underwent surgery to attach a coil support that allowed for consistent rTMS stimulation of the awake animals. Details about the surgical procedure can be found in Poh et al., 2018 [27]. In brief, animals were anesthetized with 3% isoflurane and maintained at 2–3% with a nose cone. After shaving and cleaning the site for sterility, a sagittal incision of approximately 10 mm in length was made with a scalpel blade. The periosteum was gently scraped away, and a premade coil support was attached at bregma with cyanoacrylate and reinforced with dental cement. The wound was then sutured around the coil support.

2.3. Behavior and stimulation procedures

For an overview of the behavior and stimulation procedures see Fig. 1a. After five days of recovery from the coil support surgery, the mice were habituated to a separate stimulation cage and coil procedures for three days. On the final day of coil habituation mice underwent the Open Field Test (OFT) for 10-min, which also served as the habituation for the Object Recognition Test (ORT). The following day the mice were exposed to ORT Sampling phase with two of the same objects for 10-min. Objects were of similar size but with different colors and shapes. These objects were placed in opposite corners of a square testing arena. Three hours later the mice were randomly divided into two treatment groups, TMS which received active stimulation of 10-min of 10 Hz rTMS and SHAM, which were treated the same except with the coil turned off during stimulation. Both TMS and SHAM mice were treated again 24- and 48-h post-familiarization. On the final day (72-h post-familiarization) the mice were exposed ORT Test phase for 10-min with one familiar object and one novel object. To determine if exploration of the novel object was greater than the familiar object we calculated the discrimination index (((novel object exploration)-(familiar object exploration))/((novel object exploration)+(familiar object exploration))). We chose 72-h, as both previous studies [28] and pilot data (not shown) demonstrated that under normal condition mice do not retain the ORT memory at this time point. Both the object types and the location of the novel object were balanced amongst the groups. For the habituation, sampling and probe session the mice were always given 30-min to acclimate to the experiment room before trials and, objects and the arena were cleaned with 70% ethanol after each mouse. To investigate the effects of solely the behavioral task we also included a group which were treated the same as the SHAM mice except they did not do the ORT. Instead they were exposed to a sampling and probe session but with no objects. One mouse in the TMS group did not reach a required threshold of a total 20-s exploration of both objects in the sampling session and therefore was excluded from the study. This mouse was excluded from subsequent biochemical analysis. All other mice reached criteria in both the sampling and probe session. Excluding the one mouse, the final numbers in the experimental groups were TMS n = 10, SHAM n = 11 and CONTROL n = 11.

2.4. Western blot

Three hours following the probe session the mice were sacrificed with ketamine overdose, and the hippocampus and frontal cortex tissues were rapidly dissected onto dry ice and stored at −80 °C. These samples were homogenized with 300 μL ice cold Syn-Per Synaptic Protein Extraction Reagent (Thermo Scientific) with 1x Protease Inhibitor Cocktail (Roche) and 1x Phosphatase Inhibitor Cocktail (Roche) tablets (per 10 ml), for 5 s. Homogenates were centrifuged at 1200 g for 10-min to remove cell debris. The supernatant was taken, and a portion of the supernatant was aliquoted for analysis of the Full Homogenate. The remainder was centrifuged at 15,000 g for 20-min at 4 °C. The pellets, containing synaptosomes, were gently re-suspended in Syn-Per at 1.5 ml/g of tissue, to create the Synaptic Homogenate. A nano drop was used to quantify protein concentration in the sample. The Full Homogenate and Synaptic Homogenate were stored at −80 °C until analysis.

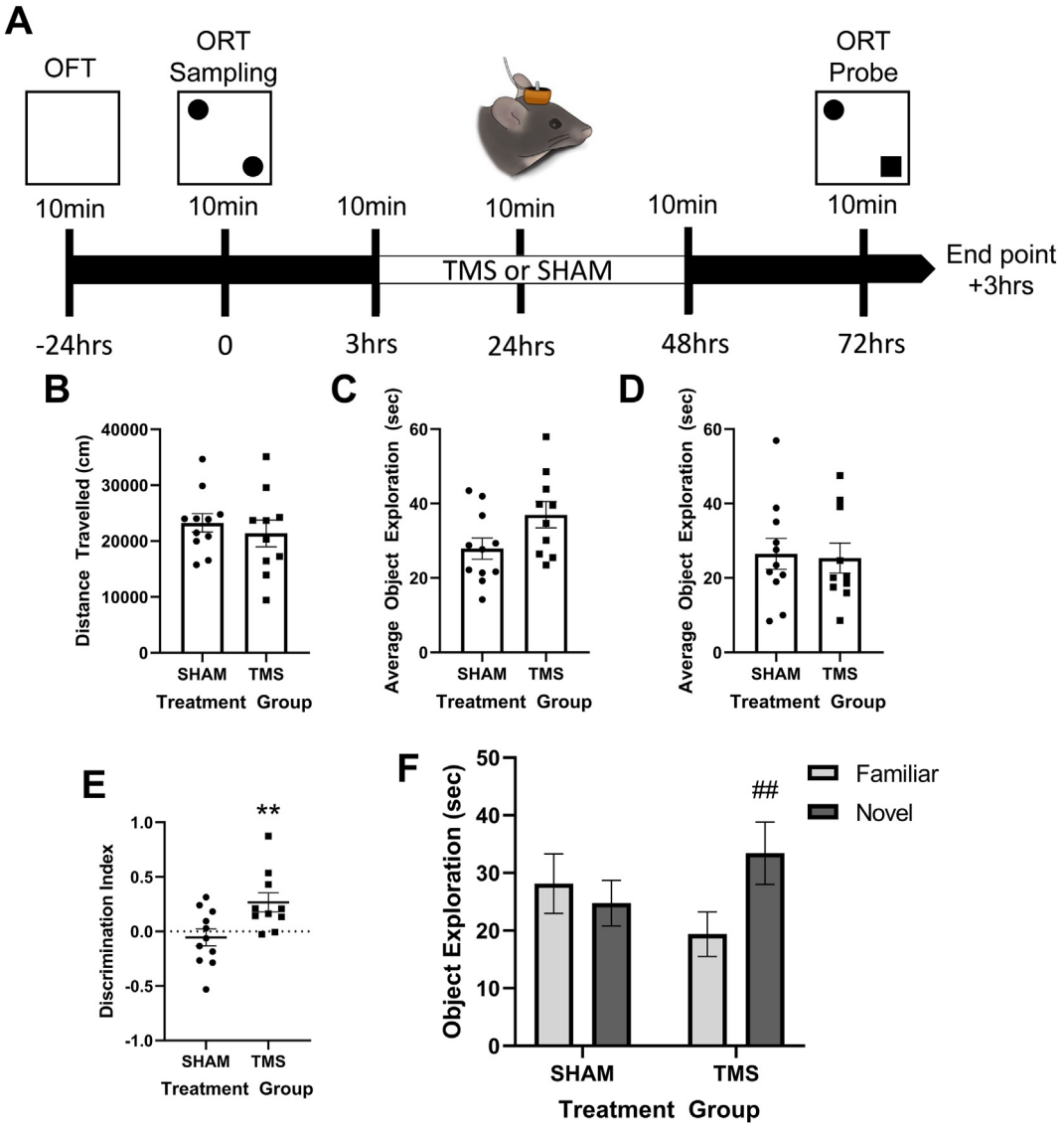


Fig. 1. rTMS Improves Consolidation of Object Recognition Test (ORT). A) Timeline of behavior and stimulation events. One day before the ORT the mice were habituated to the testing arena with the Open Field Test (OFT). The following day they underwent the ORT sampling phase where they were exposed to two of the same objects for 10 min. The mice were then treated with rTMS or SHAM for three stimulation sessions, +3, +24 and 48 h post sampling phase. The next day the mice underwent the ORT probe phase with one familiar and one novel object B) There were no differences in the distanced travelled in the OFT. Average object exploration (regardless of location or type) for the C) sampling and D) probe phases of the ORT. E) The novel object recognition index and F) comparing within groups Novel vs Familiar object exploration revealed a better memory performance in the mice which received the TMS. All figures show mean ± SEM. **p < 0.01 vs SHAM, ##p < 0.01 vs Familiar Object.

Both Full and Synaptic homogenates were diluted to 4 μg/ml in ice-cold PBS and mixed with 1:4 Protein Loading Buffer (LiCor) and 1:4000 mercaptoethanol. They were then heated at 95 °C for 5-min and loaded into Mini PROTEAN TGX Gels (Bio-Rad). After separation using electrophoresis at 100 V, the proteins were transferred onto nitrocellulose blotting membrane (GE Healthcare Life-sciences). The membranes were washed in SuperSignal Western Blot Enhancer Antigen Pretreatment Solution (Thermo Scientific) and then blocked with Blocking Buffer (Rockland) and 1% goat serum for 1-h. Following blocking, the membrane was incubated with primary antibodies diluted in SuperSignal Western Blot Enhancer Primary Antibody Diluent (Thermo Scientific) overnight at room temperature. Primary antibodies used were mouse monoclonal ERK 1/2 (1:1000, Santa Cruz Biotechnology), rabbit monoclonal phospho-p44/42 MAPK (pERK 1:1000, Cell Signaling Technologies), mouse monoclonal CREB (1:500, Cell Signaling Technologies), rabbit monoclonal phospho-CREB (S133) (pCREB 1:500, Cell Signaling Technologies), rabbit monoclonal CAMKII-alpha (1:1000, Cell Signaling Technologies), mouse monoclonal phospho-CAMKII (Thr286) (pCAMKII 1:1000, Invitrogen), mouse monoclonal GluR1 (1:750, Santa Cruz Biotechnology), rabbit monoclonal phospho-GluA1 (S845) (pGluR1(845) 1:750, Cell Signaling Technologies), rabbit monoclonal phospho-GluA1 (S831) (pGluR1(831) 1:750, Cell Signaling Technologies), mouse monoclonal GluR2 (1:750, Invitrogen), rabbit polyclonal phospho-GluR2 (S880) (pGluR2 1:500, Invitrogen), mouse monoclonal TrkB (1:500, BD Transduction Laboratories) or rabbit polyclonal phospho-TrkB (Tyr816) (pTrkB 1:500, EMD Millipore). Rabbit polyclonal Cofilin (1:2000, Sigma) was used for a standard loading control. After rinsing in TBS-Tween, the membrane was incubated for 1 h at room temperature in goat anti-rabbit 800 and goat anti-mouse 680 (1:3000, LiCor). The immunoblot was imaged using Odyssey Clx

Infrared imaging system (LiCor) and data analysis of band intensity performed using Image Studio (LiCor).

2.5. Immunohistochemistry

Three hours following the probe session, mice sacrificed with ketamine overdose, were then transcardially perfused with 50 ml of cold 1 M PBS. Brains were immediately postfixed in 4% paraformaldehyde solution at 4 °C and transferred to 30% sucrose in PBS at least 48-h before cryosectioning into 30 μm sections. Free floating sections were blocked and permeabilized with 10% goat donkey serum in 0.1% Triton in 1 M PBS then incubated for 48-h with the rabbit polyclonal PSD95 (1:300, Thermo Scientific) and mouse monoclonal SV2A (1:300, Santa Cruz Biotechnology). The sections were then incubated for 1-h with the secondary antibodies Alexa Fluor goat anti-rabbit 568 (1:500, Invitrogen) and Alexa Fluor goat anti-mouse 488 (1:500, Invitrogen. The slices were then washed with nuclear dye DAPI (1:1000, Sigma), mounted onto slides and cover slipped with Fluoromount (Thermo Scientific). Staining was quantified in the frontal cortex (FC), dorsal CA1 hippocampus, entorhinal cortex (EC) and perirhinal cortex (PC). Z-stacked images were taken with a BZ-X700 Fluorescent Microscope (Keyence) at 40x magnification to identify synaptic puncta throughout the entire slice. Total PDS95 puncta, SV2A puncta and colocalized (defined as two puncta next to each other or overlapping) were counted from at least four images for each area and the average of the results taken for each mouse. Quantification of % colocalized was completed by counting (overlapping SV2A and PSD95)/(total SV2A and PSD95) puncta.

2.6. Statistics

All data met assumptions of normality as assessed by examining the skewness, kurtosis and the Shapiro Wilk value of the calculated residuals and there were no violations of the homogeneity of variance as assessed using Levene's statistic. For the behavior tests, students t-tests were used to compare SHAM to TMS, for the exploration of novel and familiar objects within groups a Repeated Measures Two-Way ANVOA was used. Immunohistochemistry and Western blot data were tested for significance using a One-Way ANOVA. For all post-hoc pairwise comparisons a Tukey adjustment was used with significance at $p < 0.05$. If the homogeneity of variance was violated, significance was determined with the Welch Test and subsequent post-hoc corrected with Games-Howell, these cases are noted in the results. All statistics were performed using SPSS (version 22, IBM) and graphs produced using GraphPad (version 7, Prism).

3. Results

We used the ORT with a 72-h delay between the sampling and test phase to assess the effectiveness of rTMS on memory. There was no difference in the OFT activity and object exploration times in the sampling and test trial between SHAM v TMS ($p > 0.05$, Fig. 1B-D). Additionally, there was no bias for object location in the acquisition trial ($p > 0.05$, data not shown). The mice which received the TMS over 72-h had a significant discrimination index of the novel object (One way *t*-test $t(9) = 3.067$, $p = 0.0134$), whereas SHAM mice did not (One way *t*-test $t(10) = 0.705$, $p = 0.496$). Comparing the two discrimination indices, TMS was significantly greater than the SHAM (Student t-test $t(19) = 2.77$, $p = 0.0122$, Fig. 1E). In addition, only the group which received rTMS had a significant t- We also analyzed the difference within each treatment group of exploration of the novel and familiar object and found a significant difference between the object explored and the treatment given (Two Way Repeated Measures ANOVA $F(1, 19) = 8.305$, $p = 0.0096$, $\eta_p^2 = .304$ Fig. 1F). Post-hoc analysis showed that the TMS group had a significant increase in novel object exploration ($p = 0.0092$), but no difference in the SHAM group ($p > 0.05$), indicating that rTMS between the learning and retrieval event better consolidated the memory.

Following this result, we were interested in understanding the neurobiological changes associated with the improved memory performance in mice that received rTMS. We used Western Blot to analyze the amount of CREB, CAMKII and ERK and their phosphorylation using the Full homogenate from the hippocampus and FC (Fig. 2). In the hippocampus, overall, there was a change in the amount of CREB (One-Way ANOVA $F(2,15) = 6.20$ $p = 0.013$, $\eta_p^2 = .488$ Fig. 2a). Post-hoc analysis revealed that this was derived from an increase in CREB in both the groups which did the behavioral test, SHAM ($p = 0.048$) and TMS ($p = 0.016$) compared to the CONTROL group. However, it was the phosphorylation of CREB that appeared to be affected solely by the rTMS in the hippocampus (One-Way ANOVA $F(2,15) = 11.813$ $p = 0.001$, $\eta_p^2 = .645$ Fig. 2a), as post-hoc analysis showed the mice that received rTMS had significantly greater levels of pCREB than the CONTROL ($p = 0.001$) and the SHAM group ($p = 0.041$). Additionally, we found an effect on the levels of pCAMKII (One-Way ANOVA $F(2,15) = 4.817$ $p = 0.029$, $\eta_p^2 = .445$, Fig. 2d), where similarly to pCREB post-hoc testing showed that rTMS increased the amount of pCAMKII compared to the CONTROL group ($p = 0.027$).

In the frontal cortex, the phosphorylation of CREB that appeared to be affected by the rTMS (One-Way ANOVA $F(2,15) = 10.572$ $p = 0.003$, $\eta_p^2 = .658$, Fig. 2b). Post-hoc analysis showed the mice that received rTMS had significantly greater levels of pCREB than the CONTROL group ($p = 0.002$)/There was also an effect on pCAMKII in the FC (One-Way ANOVA $F(2,15) = 6.452$ $p = 0.011$, $\eta_p^2 = .498$, Fig. 2e), however post-hoc analysis showed that it was regardless of the treatment as both SHAM ($p = 0.016$) and TMS (0.035) groups had increased pCAMKII compared to the CONTROL group. In the FC the increased pCAMKII was due to an increased phosphorylation of existing CAMKII, as the proportion of pCAMKII to CAMKII was also increased with the behavioral test (One-Way ANOVA $F(2,15) = 7.387$ $p = 0.007$, $\eta_p^2 = .532$, Fig. 2e). Post-hoc testing showed that both SHAM ($p = 0.026$) and TMS (0.010) had a significantly greater portion than CONTROL. In both the hippocampus and FC we did not see any significant changes in ERK or its phosphorylation after rTMS.

We used Western Blot on the Synaptic homogenates from the hippocampus and FC to determine the effect of rTMS on different synaptic receptors involved in learning and memory. We examined the change in AMPA receptor subunits GluR1 and GluR2, as well as the BDNF receptor TrkB. In the hippocampus there was a difference in the levels GluR2 (Welch Test $F(2,15) = 6.718$ $p < 0.001$, Fig. 3g) and pGluR2 (One-Way ANOVA $F(2,15) = 7.189$ $p = 0.023$, $\eta_p^2 = .461$, Fig. 3g). Post-hoc analysis revealed SHAM mice had less GLUR2 than the CONTROLS (Games-Howell $p = 0.022$) and the mice which received rTMS treatment had significantly higher levels of both GluR2 (Games-Howell $p = 0.037$) and pGluR2 ($p < 0.05$) than the SHAM mice. In the hippocampus we also found changes to pTrKB (One-Way ANOVA $F(2,15) = 8.449$ $p = 0.005$, $\eta_p^2 = .585$, Fig. 3J). As post-hoc testing showed both the SHAM mice ($p = 0.008$) and TMS mice ($p = 0.013$) had less pTRKb than CONTROL it suggests this effect was related to the behavior test and independent of treatment. This effect was also due to change in the proportion of total TrkB that was phosphorylated (One-Way ANOVA $F(2,15) = 8.882$ $p = 0.004$, $\eta_p^2 = .597$, Fig. 3J), both SHAM ($p = 0.042$) and TMS ($p = 0.004$) were significantly lower than CONTROL.

The only effect on GluR1 was found in the FC where there was a change in the proportion of GluR1 phosphorylated at the S845 site,

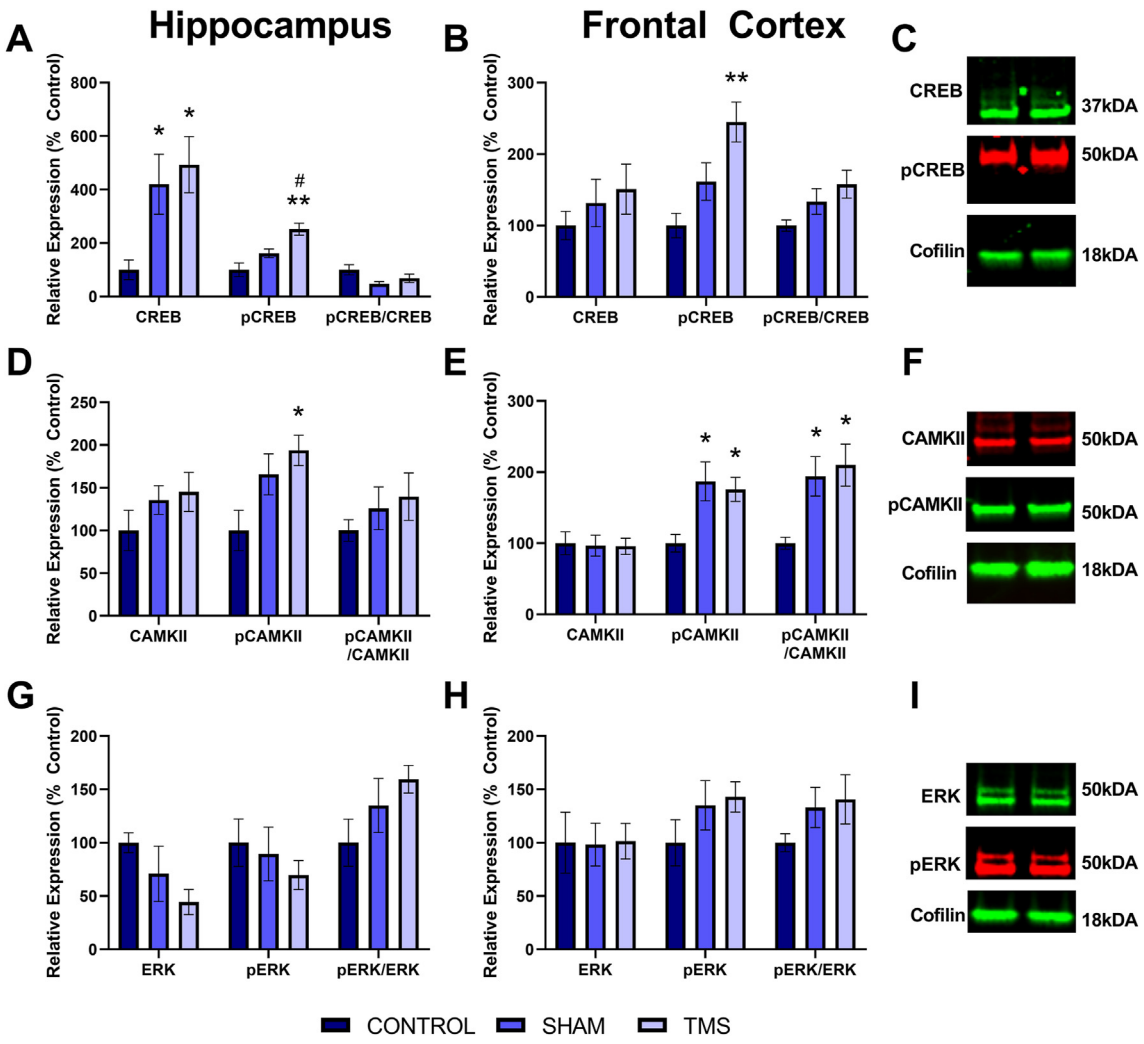


Fig. 2. rTMS Effects the Response of Synaptic Signalling Proteins. Mean (+/− SEM) protein expression in the Hippocampus and Frontal Cortex as measured with western blot of A-C) CREB and pCREB, D-F) CAMKII and pCAMKII and G-I) ERK and pERK. Overall expression and percentage phosphorylated is shown. *p < 0.05 **p < 0.01 vs Control, #p < 0.05 vs SHAM.

pGluR1(845) (One-Way ANOVA $F(2,15) = 4.170$ $p = 0.040$, $\eta_p^2 = .391$, Fig. 3e), but not the S831 site. Post-hoc analysis showed that there was a decrease after rTMS compared to CONTROLS ($p = 0.038$). There was no change in GluR2 receptors in the FC however, we found an effect on the levels of TrkB (One-Way ANOVA $F(2,15) = 13.601$ $p = 0.001$, $\eta_p^2 = .712$, Fig. 3K) and pTrkB (Welch Test $F(2,15) = 4.745$ $p = 0.035$, Fig. 3K). Post-hoc analysis showed that this effect was only due to an increase in levels after rTMS treatment compared to controls for both TrkB ($p = 0.001$) and pTrKB (Games-Howell $p = 0.037$). As there was no difference in the proportion of pTrKb to TrkB (Fig. 3K) this suggests that unlike in the hippocampus, the change in pTrkB was resultant of the overall increase in TrKB.

Finally, we investigated the effect of rTMS stimulation on increasing and maintaining synaptic connections in areas of the brain that are involved in the ORT. We used immunohistochemistry to determine colocalized presynaptic (SV2A) and postsynaptic (PSD95) puncta to indicate the proportion of active, connected synapses in these areas. Overall, there was change in the number of colocalized synapses in the CA1 region of the hippocampus (One-Way ANOVA $F(2,16) = 4.231$ $p = 0.038$, $\eta_p^2 = .377$, Fig. 4a), the EC (One-Way ANOVA $F(2,16) = 4.658$ $p = 0.030$, $\eta_p^2 = .392$, Fig. 4c) and the PC (One-Way ANOVA $F(2,16) = 7.895$, $p = 0.006$, $\eta_p^2 = .548$, Fig. 4d). In both the CA1 region and PC there was increased colocalization in the TMS group to CONTROL (CA1 $p = 0.031$, PC $p = 0.005$), however in the EC there was an increase from CONTROL in both the SHAM ($p = 0.047$) and TMS ($p = 0.022$) groups, suggesting an overall effect of the behavior test in this region with little effect of the treatment. We did not see any significant changes in the FC.

4. Discussion

Our results show that rTMS, if given between encoding and retrieval, improves memory in the ORT in mice. These results reflect human studies which have shown a similar effect of post-encoding rTMS or other forms of non-invasive brain stimulation, on improving memory [2,3]. Predominantly rodent studies have focused on pre-encoding rTMS for improving memory in the past [29–32], therefore the neurobiological mechanisms of consolidation targeted rTMS, even though for specific memory improvement it is more effective than other stimulation time points [1], have not been investigated.

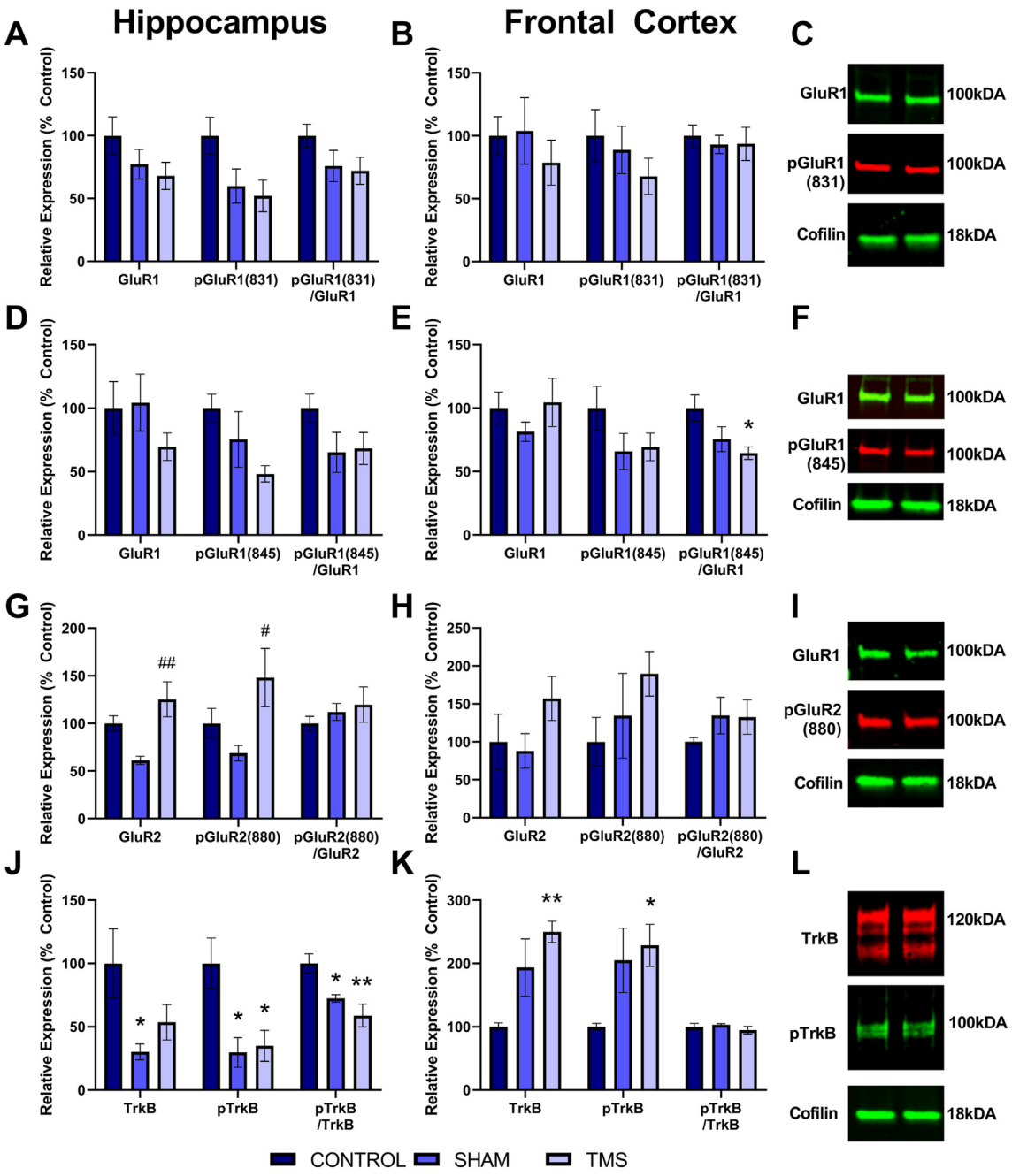


Fig. 3. rTMS Changes Receptor Dynamics at the Synapse. Mean (+/− SEM) synaptic protein expression in the Hippocampus and Frontal Cortex of A-C) GluR1 and pGluR1(831), D-F) GluR1 and pGluR1(845), G-I) GluR2 and pGluR2(880) and J-L) TrkB and pTrkB. Overall expression and percentage phosphorylated is shown. *p < 0.05 **p < 0.01 ***p < 0.001 vs Control, #p < 0.05 vs SHAM, ##p < 0.01 vs SHAM.

An indicator of persistent LTM is the formation of strong, stable synaptic connections. In both the PC and CA1 only the mice which had rTMS had increased synaptic connections compared to the CONTROL mice, whilst SHAM mice did not. The PC and dorsal CA1 are important in object recognition memory through distinct mechanisms [33]. While the PC is involved in the recognition of objects based on their novelty or familiarity, the hippocampus is involved in the encoding the specific object information from sensory stimuli into memory [33–35]. As both regions had elevated synaptic connections in the rTMS mice, which had improved memory, this suggests that the stimulation was maintaining both of the important connections in these areas that encode both the object information and familiarity.

One of the most striking biological changes we observed was the increase in the expression of GluR2 at the synapse of the hippocampus after rTMS treatment improved memory compared to a decrease in the SHAM mice. This was in contrast with very little change to GluR1 expression. Previous studies have shown that GluR2 receptors are important for LTM storage and the removal of GluR2 receptors from the synapse is involved in forgetting [36]. Contrarily, blocking GluR2 endocytosis prevents forgetting in experimental settings [37]. There have only been a few studies that have shown conflicting evidence to the effect of rTMS on GluR2

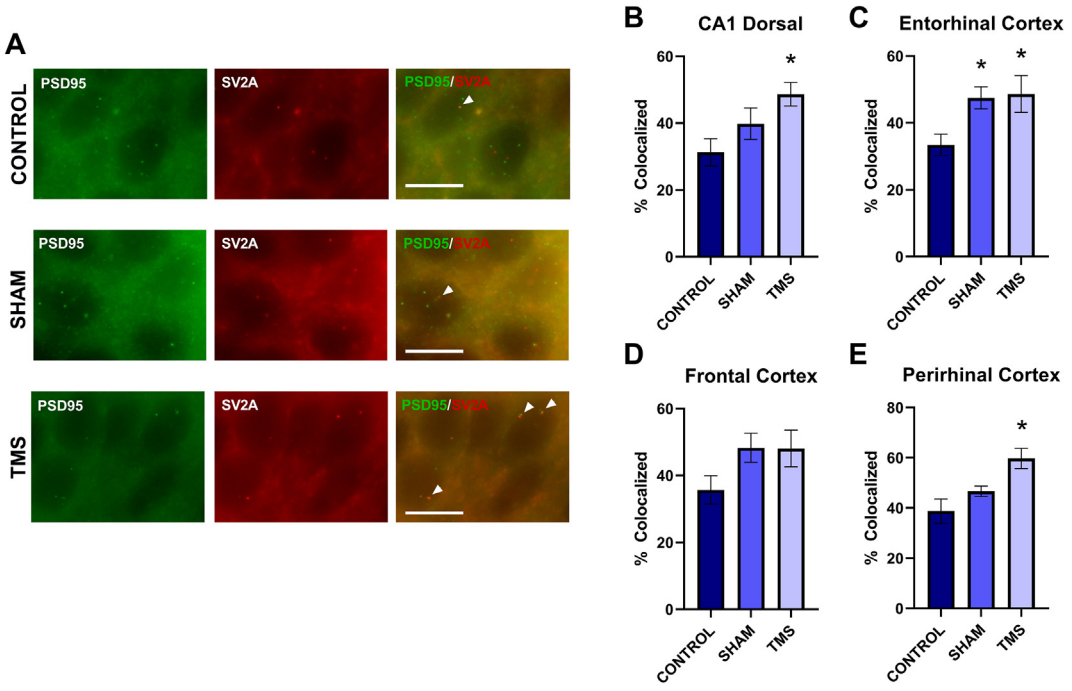


Fig. 4. rTMS Increases Region Specific Synaptic Connections. A) Example of PSD95 (green), SV2A (red) and colocalized (white arrow) puncta in the dorsal CA1 region of the hippocampus for a CONTROL, SHAM and TMS mouse (scale bar 10 μm). Mean (+/− SEM) of the percentage of total puncta which had colocalized SV2A and PSD95 in the B) Dorsal CA1 region of the hippocampus, C) Frontal Cortex, D) Entorhinal Cortex and E) Perirhinal Cortex. *p < 0.05 vs CONTROL.

receptors [26,38,39]. We believe our results show a unique effect as the rTMS was paired with the learning task which resulted in the forgetting and loss of GluR2 receptors in the SHAM mice. We therefore have shown for the first time that rTMS can promote the stability of GluR2 receptors at the synapse in relation to the object recognition task, which as demonstrated in previous literature, is important to maintain a memory [36,37].

Interestingly in the FC, rTMS was associated with an increase in TrkB and the phosphorylation of TrKB, whereas in the hippocampus, the memory task reduced pTrkB expression. The kinetics of TrkB at the synapse are dependent on many factors, as TrkB can be internalized at the synapse once BDNF binds [40]. However, longer activation periods of BDNF result in a stable upregulation of TrkB whereas transient BDNF increases TrkB only temporarily [41]. Therefore, the differing expression of FC and hippocampus TrkB may be an indication of region-specific effects of BDNF. In the hippocampus TrkB internalization may have occurred in both the SHAM and TMS mice in response to the behavioral test, as conducting the test phase may have caused some level of reconsolidation learning event in both of these groups, which would have activated a short BDNF response [42–44]. Whereas in the FC, as previous studies have shown direct increases in BDNF from rTMS [18–20], the persistent increase in BDNF from the stimulation over the three days may have promoted the upregulation and translocation of TrkB to the synapse [41]. Additionally, the effect of the behavior test causing TrkB internalization would not have occurred in the FC as it did in the hippocampus, as the FC is not involved in recognition memory [45–47]. This was also demonstrated in our study as there was no change in synaptic connections in this region. Future studies should explore rTMS versus behavioral test effects in more detail to determine this exact mechanism.

From our results there appears to be a strong link between CREB and the improved memory performance in the mice which received rTMS. Overall, we saw elevated levels of CREB and its phosphorylation in both the hippocampus and FC. This result can be related to previous research which has also shown the ability for rTMS to increase CREB expression and its phosphorylation [20,21]. Increased pCREB is important for the transcriptional activity needed to convert a short term memory to a stable LTM [48–50] and blocking CREB prevents consolidation of memory in the hours following the learning event [51]. Therefore, at the consolidation stage, it would be important if rTMS could increase pCREB. However, the role of the prolonged activation of CREB is less clear. This study shows that phosphorylation of CREB was associated with improved memory and therefore may be important in the improved memory performance we saw in the rTMS treated mice.

CREB activation can come from many synaptic pathways. Unfortunately, this study did not determine the exact pathway that would result in elevated pCREB. The phosphorylation of CAMKII was increased in both the hippocampus and FC which could cause the increased CREB phosphorylation [9–11]. As rTMS alters Ca^{2+} levels within neurons [18], this activation of CAMKII may have provided a direct pathway to activate CREB, additionally CAMKII activation has been directly linked to GluR2 activity [52]. However, in the FC the SHAM mice also had elevated CAMKII but not CREB or GluR2. The synaptic staining revealed that the hippocampus plays a greater role in object recognition memory than the FC, so there may be other regulatory pathways involved which supported the activation of CREB from CAMKII after rTMS in the hippocampus and not the FC.

From this study it is difficult to distinguish whether the improvements from rTMS were solely due to the stimulation given 3 h post-encoding or the stimulations given within the 72-h. Singular consolidating stimuli are effective in improving memory in human trials [3], however in rodent studies consistent reactivation of memory pathways is essential for LTM maintenance [17]. Additionally, from the results in this study we cannot identify whether the improved memory or neurobiochemical changes were result of the rTMS stimulation given during the consolidation phase improving the consolidation of the memory or if there was a

secondary mechanism altering the retrieval of the memory independent to consolidation. The results from this study show both short-term changes to phosphorylation of proteins needed from synaptic consolidation and long-term changes to stable synaptic connections needed for memory maintenance. Therefore, it will be important for future studies to compare the effect of different stimulation time points and non-probe trial control groups to determine the exact mechanisms.

5. Conclusion

Together the results suggest that by promoting greater synaptic GluR2 and preventing it's endocytosis, as well as maintaining synaptic connections in the PC and hippocampal CA1, rTMS can improve memory in the object recognition test. We found evidence that these changes were possibly linked to by CAMKII and CREB pathways in hippocampal neurons. These results further solidify delivering rTMS in the consolidation phase for LTM and altering synaptic plasticity, which could have greater implications in the clinical neuromodulation field, not only for improving memory but also in cases where a greater retention of information is beneficial, for example psychotherapy [53]. Future studies could employ similar methods to compare the consolidation and maintenance stimulation to directly determine the molecular mechanism contributing to the effect, as well as determine if this effect could be extended with prolonged maintenance rTMS.

CRediT authorship contribution statement

A.M. Heath: Conceptualization, Formal analysis, Investigation, Writing – original draft. **M. Brewer:** Investigation. **J. Yesavage:** Supervision, Writing – review & editing. **M.W. McNerney:** Supervision, Resources, Funding acquisition, Writing – review & editing.

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